W3WI_110.2 - Distributed Systems

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Version: 24SCB - May, 17 2026

Core Topics

- Terminology, concepts, architectures, requirements profiles and architecture models for distributed systems
- Design and implementation approaches
- Comparison of different middleware concepts
- Synchronous and asynchronous communication, remote method calls
- Asynchronous communication and messaging systems
- Security aspects in distributed systems

Previous Knowledge

- Who is familiar with Java?
- Who is familiar with Python?

Examination - Portfolio

Background

- the module *Developing Distributed Systems* has 55 lecture hours
- the lecture **Distributed Systems has 22 lecture hours**

  ⇒ Distributed Systems will contribute up to **50** points (out of 120) to the final grade for the module. (Please, don't do the math.)

2 Parts

01 Presentations

02 Programming Exercise

Presentations - General Conditions

- Each person should present for 10 Minutes sharp!
- The presentations should deal in particular with the core content of the lecture and *be of a conceptual nature*.

  I. e., briefly describe the underlying problem before you discuss, e.g., the algorithm/technology and how it solves the specific problem, how errors are dealt with, which services are offered, which guarantees/security aspects are implemented or how scalability is achieved.

  No promotional presentations and no extended motivation presentations. Focus on the topic at hand.
- Presentations have to be in English.

Presentations - Submissions

⚠ Attention!

You have to upload your presentation to Moodle up until 7 o'clock on the day of your presentation.

⚠ Attention!

Further Requirements and Checklist (in Ger.)

The checklist has to be signed and uploaded together with the presentation. (As a second PDF file.)

Presentations - Available Topics

Hint

Students giving presentations belonging to the same block have to coordinate with each other to avoid any overlap. If you need specific knowledge, which is expected to be covered by another student, but you are not sure whether it will be presented sufficiently, create a backup slide for your presentation that covers this topic as well and mark it as a backup slide. This backup slide will not be counted towards the time limit when you give your presentation.

For some topics, we provide some *initial ideas* regarding the potential content of the presentation. These are just suggestions and you are free to deviate from them as long as you cover the core content of the topic and provide a conceptual presentation.

!! Important

You are also not required to discuss all topics mentioned in the initial ideas. It is your job to determine the key points and provide a coherent and meaningful presentation.

Virtualization and Virtualization Platforms

01 Introduction to Virtualization & Use Cases

- What is virtualization? Historical context and motivation
- Different types/levels of virtualization
- Key use cases: server consolidation, cloud computing, development/testing, isolation, ...
- Benefits and trade-offs

02 Hypervisors - Architecture & Types

- What is a hypervisor?
- Type 1 (bare-metal) vs Type 2 (hosted) - architectural differences
- Full virtualization vs paravirtualization approaches
- Examples and when to use each type

03 Virtual Machines - Implementation & Management

- VM structure and components (virtual hardware, guest OS)
- VM snapshots
- VM migration (live migration concepts)
- Resource allocation and isolation

04 Containers & OS-level Virtualization

- Container concept and how it differs from VMs
- Namespaces and cgroups (conceptual)
- Container images and layering
- Use cases and comparison with VMs

05 Memory Virtualization

- The address translation problem (guest virtual → guest physical → host physical)
- Shadow page tables approach
- Hardware-assisted virtualization (EPT/NPT)
- Memory management techniques (overcommitment, ballooning)

- **06** Network & I/O Virtualization
  - Challenges of virtualizing network and I/O devices
  - Emulated vs paravirtualized devices
  - SR-IOV (Single Root I/O Virtualization) and device passthrough
  - Virtual NICs and network bridges
  - Virtual switches and network isolation

Network Protocols

- **01** QUIC (only available when we have ≥ 21 students)
- **02** HTTP/3
- **03** BitTorrent Protocol (only available when we have ≥ 25 students)

Modern RPC

- **01** Protobuf
- **02** Google RPC

Web-App Security

- **01** SOP (Same-Origin Policy), CORS (Cross-Origin Resource Sharing) (Foundations)
- **02** CORP / COOP / COEP (Cross-Origin Resource/Opener/Embedder Policies) (only available when we have ≥ 23 students)
- **03** CSP (Content Security Policy) and SRI (Subresource Integrity)

Introduction and concrete examples how they are used/specified and help prevent attacks.

Monitoring & Debugging Distributed Systems

- **01** Log Aggregation with a particular focus on correlation of log entrie collected from different nodes in a distributed system.

Leader Election

- **01** Bully Algorithm and Ring Algorithm

Quorum Systems

- **01** Majority voting (i. e., quorum-distributed computing)

Consensus Algorithms and Fault Tolerance

- **01** Consensus Fundamentals & Problem Definition
  - What is consensus and why is it hard in distributed systems?
  - The FLP impossibility result (conceptual understanding)
  - Fault models: crash faults vs Byzantine faults
  - Safety vs liveness properties
  - Real-world motivation: replicated state machines, distributed databases
- **02** (Practical) Byzantine Fault Tolerance
  - When do we need BFT?
  - Modern developments

- Real-world usage

03 Paxos Family

- Basic Paxos algorithm (conceptual overview, roles: proposers, acceptors, learners)
- Why Paxos is correct but complex
- Multi-Paxos for practical systems
- Real-world usage

04 Raft - Understandable Consensus

- Leader election, log replication, safety
- How Raft differs from Paxos (design philosophy)
- Real-world usage

Eventual Consistency

01 Eventual Consistency and Gossip Protocol

02 CRDTs (Conflict-free Replicated Data Types) (only available when we have ≥ 22 students)

Distributed File Systems

01 Ceph

02 HDFS (only available when we have ≥ 24 students)

Topic Assignment

The presentations need to be given in the order listed below.

Topic	Constraint	Student Name
01. Introduction to Virtualization & Use Cases		
02. Hypervisors - Architecture & Types		
03. Virtual Machines		
04. Containers & OS-level Virtualization		
05. Memory Virtualization		
06. Network & I/O Virtualization		
07. QUIC	≥ 21 students	
08. HTTP/3		
09. BitTorrent Protocol	≥ 25 students	
10. Protobuf		
11. Google RPC		
12. SOP, CORS (Foundations)		
13. CORP / COOP / COEP	≥ 23 students	
14. CSP and SRI		
15. Log Aggregation		
16. Leader Election: (a) Bully, (b) Ring Algorithm		
17. Quorum Systems (Majority Voting)		
18. Consensus Fundamentals & Problem Definition		
19. Byzantine Fault Tolerance (PBFT)		
20. Paxos Family		
21. Raft - Understandable Consensus		
22. Eventual Consistency and Gossip Protocol		
23. CRDTs	≥ 22 students	
24. Ceph		
25. HDFS	≥ 24 students	

Presentations - Grading (max 20 Points)

- Presentation (max 17 Points)
- Checklist (max 1 Point)
- Moderation (max 2 Points)

  A Moderator has three main tasks:

  1. (Introduce the speaker) and *motivate the topic*.
  2. Keep track of time and *give time warnings to the presenter* (2 minutes left, time is up); abort the presentation if necessary (-1 minute).
  3. Lead a short Q&A session (2-3 questions) after the presentation; if there are no questions from the audience, *the moderator should have two to three questions* that the presenter is not aware of in advance (e.g., by asking the presenter for potential questions related to the topic beforehand).

Programming Task (max 30 Points)

- Carried out in Groups of two students (and one group with three students if we have an odd number of students)
- You'll get 1 point for a correct upload of the required files.
- You'll get 1 point if the public test suite runs successfully on your solution.
- You'll get up to 8 points based on the results of running an extended test suite on your solution. The grading will be based on the number of test cases passed successfully.
- Grading will be done during an examination interview (20 Points) (15 Minutes per Group):
  1. Demonstration of your project (~5 Minutes)
  2. Answering specific questions related to your implementation
- The download will be made available in due time.

Lecture - Schedule

May 18, 2026 (5VL):
- Lecture

June 3, 2026 (2VL):
- Lecture
- Programming Task Explanation

June 16, 2026 (5VL):
- Presentations:
  - 01. Introduction to Virtualization & Use Cases (Baha Sahin)
  - 02. Hypervisors - Architecture & Types (Thorge Hartmann)
  - 03. Virtual Machines (Jan Wendlandt)
  - 04. Containers & OS-level Virtualization (Heba Alkhaled)
  - 05. Memory Virtualization (Jakob Lindqvist)
  - 06. Network & I/O Virtualization (Can Sever)
  - 07. QUIC (Simon Wagner)
  - 08. HTTP/3 (Yuxi Yan)
- Lecture

June 22, 2026 (5VL):
- Presentations:
  - 10. Protobuf (Mick Burcksen)
  - 11. Google RPC (Luisa Homola)
  - 12. SOP, CORS (Foundations) (Mia Roth)
  - 13. CORP / COOP / COEP (Eliano-Markus Glock)
  - 14. CSP and SRI (Giulia Petry)
  - 15. Log Aggregation (Omid Saidi)
  - 16. Leader Election: Bully and Ring Algorithm (Johanna Jansen)
  - 17. Quorum Systems (Majority Voting) (Armin Mirzakhanian)
- Lecture

30 June, 2026 (5VL):
- Presentations:
  - 18. Consensus Fundamentals & Problem Definition (Jonathan Lahmer)
  - 19. Byzantine Fault Tolerance (PBFT) (Simon John)
  - 20. Paxos Family (Janis Josefowski-Pätz)
  - 21. Raft - Understandable Consensus (Daria Makaschin)
  - 22. Eventual Consistency and Gossip Protocol (Aleksander Orlowski)
  - 23. CRDTs (Luca Santoro)
  - 24. Ceph (Leni-Marie Sternberger)
  - 25. HDFS (Stephan Fürstenau)
- Lecture

July 27 2026, 07:00 (7:00 a.m.):

- Programming Task Submission Deadline

  Uploaded by *one team member* only:

  - the two JAR files as described in the task description
  - a ZIP file with the source code
  - a PDF file which explicitly states who worked on which part and which (optionally!) states whether you want to be graded as a group or individually w.r.t the submitted code. The oral exam is always graded individually.

July 28, 2026:

- Oral Examination (Programming Task)

  Duration: 15 minutes

  During the oral examination, you will be asked to present, demonstrate, and explain your solution.

  The examination will focus on your solution to the programming task and the concepts directly related to it. It will not cover unrelated topics.

  You are required to bring your own notebook computer with a functioning IDE, all source code, and precompiled binaries. Your solution must be fully operational and ready to run on your device.

Exam Interviews (July, 28 2026)

13:00	Jonathan Lahmer	Thorge Hartmann
13:20	Johanna Jansen	Luisa Homola
13:40	Armin Mirzakhanian	Heba Alkhaled
14:00	Yuxi Yan	Leni-Marie Sternberger
14:20	Pause	Pause
14:30	Daria Makaschin	Stephan Fürstenau
14:50	Mia Roth	Giulia Petry
15:10	Baha Sahin	Simon John
15:30	Luca Santoro	Can Sever
15:50	Pause	Pause
16:00	Simon Wagner	Janis Josefowski-Pätz
16:20	Aleksander Olowski	Omid Saidi
16:40	Eliano-Markus Glock	Jakob Lindqvist
17:00	Jan Wendlandt	Mick Burcksen